Angular Sparsity Is an Atomic-Sector Property: The Abelian Residue of Multi-Center Bonding*

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The GeoVac program builds molecular Hamiltonians from atom-centered hydrogenic blocks (screened Z_{eff} per block, *not* shared-p_0 Coulomb–Sturmian — a design choice, *not* a theorem: Papers 8–9 establish R-independence only under the imposed single-n D-matrix premise (Paper 8, Remark "Scope: the overlap premise is imposed, not generic"), and the genuine cross-n Coulomb–Sturmian overlap escapes it and binds) whose angular matrix elements are generated from quantum-number labels by Gaunt selection, with no spatial quadrature. This yields structurally sparse qubit Hamiltonians—838 Pauli terms for LiH at $Q = 30$, exactly linear in Q across molecules at fixed basis (Paper 14)—at the cost of a documented binding failure for second-row diatomics (the W1e wall, Papers 19/20 [5, 6]).

We settle the question of whether that trade was necessary. Using exact rational two-center machinery (Mulliken auxiliary functions A_n/B_n for overlaps; a hydrogenic eigen-trick for cross-center one-body terms, certified by an exact bra/ket identity holding entrywise), we census the *genuine* bare tensor (S, h, g) for a real fragment pair and find: **[MEASURED]** the Gaunt l-selection rule does not survive cross-center evaluation, while the axial m-rule does; cross-center one-body entries reach 0.80 Ha—bond scale, not perturbative. **[COUNTED]** the symmetry-*permitted* electron-repulsion density inflates 13.8× over the builder at $n_{\max} = 2$ and 15.0× at $n_{\max} = 3$; the $n_{\max} = 2$ figure is independently corroborated numerically (29.8% measured against 29.4% counted), the $n_{\max} = 3$ figure is counted only.

We give the mechanism **[INTERNAL THEOREM]**: m labels the *abelian* symmetry shared by both centers (rotation about the internuclear axis is one global one-parameter group, so m is a common label), while l labels irreducible representations of a rotation group attached to a single center. The intersection of the two centers' rotation groups is exactly the axial $U(1)$, whose character group is $\mathbb{Z}$ by Pontryagin duality. Abelian selection rules are additive charge conservation and survive; non-abelian Clebsch–Gordan selection requires the full sphere and does not. The surviving *cross-center* angular rule is therefore the one generated by the common axial subgroup, which is readable from the molecular point group in advance.

Two scope statements, since the title is shorter than the claim. "Abelian residue" names what survives *cross-center among the continuous per-center labels*—not everything that survives: same-center Gaunt selection persists from the full per-center $SO(3)$, and the largest group fixing both nuclei is $C_{\infty v}$, which is non-abelian and does yield usable label structure (its σ_v reflection is the $\mathbb{Z}_2$ we use as the tapering baseline). And the *mechanism* is classical, not ours: the intersection rule is Curie's principle (1894), the reading of surviving labels as residual-subgroup irreducibles is descent in symmetry after Bethe (1929), and the character step is elementary Pontryagin duality. What is new here is the *target*— symmetry descent aimed at the sparsity of the two-center integral tensor rather than at spectroscopic level splittings—together with the census that quantifies it.

Two re-readings follow. First, the 17.9× Pauli inflation previously attributed to Löwdin orthogonalization (v4.73.1) was not, in the bulk, orthogonalization's price: the genuine tensor's own permitted density inflates by the same order ($\sim 15\times$) before any transformation, so Löwdin was paying a bill that was largely already owed. Second **[OBSERVATION]**, four previously attempted relocations of the non-orthogonality—into the operator, into a dual basis, into a high-symmetry embedding, into device-measured many-body overlaps—each moved the cost without reducing it, because the cost measures the physical asymmetry of a two-center problem. A fifth, measured here, is the sharpest: enabling equivalent-atom-swap tapering *reduces* the qubit saving (BeH_2 12 → 9, N_2 22 → 19, F_2 22 → 15) while inflating the Pauli count 2.3–3.6×, because the swap rotation merges per-block abelian gradings into joint stabilizers. A permutation identifies blocks, and identifying blocks destroys the per-block labels that generate the sparsity.

We close with a demonstration **[PANEL-VERIFIED]**: all-electron 12-electron NaH, the framework's hardest documented failure, binds at $R_{\text{eq}} = 3.736\, a_0$ (+4.8% vs experiment; the coarse-grid dissociation-scan well minimum, now test-backed) with 91.1% of in-basis FCI binding from three fragment-native determinants, once the integrals are genuine end-to-end—with *fewer* correlation degrees of freedom than the runs that failed, which is what makes the inference go through. This localizes W1e to the Hamiltonian specification rather than the correlation treatment on this system. The demonstration deliberately forfeits the sparsity: it makes no sparsity claim.

[SYMBOLIC + MEASURED] Separately, the diatomic integrals are made *quadrature-free*: the minimal-basis H_2 FCI energy is certified to 60 digits with symbolic τ-series termination, its entire potential curve $E(R)$ is a single closed-form expression in $\{\exp, E_1, \log, \gamma\}$ with certified R_{eq},

D_e and k, and the heteronuclear LiH case reaches 30 digits net of an explicit tail bound.

A by-product bears on the continuous theory the census borrows from: the radial integral at the core of the two-center direct-Coulomb $(AA|BB)$ repulsion closes in elementary form (weight one, π-free, no dilogarithm), so that block is not merely counted but *decided*—all 195 permitted entries proven nonzero by Lindemann–Weierstrass separation.

I. INTRODUCTION: WHAT THE SPARSITY BOUGHT, AND WHAT IT COST

The GeoVac framework's computational proposition is structural sparsity. Angular matrix elements are generated from the quantum-number labels (n, l, m) by Gaunt selection rules and Wigner $3j$ symbols with no spatial integration, so the resulting second-quantized Hamiltonian is sparse by construction rather than by numerical thresholding. The downstream consequence is the qubit-resource result of Paper 14 [3]: composed LiH at 838 Pauli terms on $Q = 30$ qubits, a within-molecule basis exponent of $O(Q^{3.17})$ against $O(Q^{3.9–4.3})$ for Gaussian baselines — a narrower margin than the retired pair-diagonal figures suggested — and a 76× reduction against cc-pVDZ.

The framework's documented failure is chemical binding for second-row diatomics. Named W1e, it has resisted an unusually thorough sequence of attacks: six Phillips–Kleinman modifications, basis-level Schmidt orthogonalization, core-correlation estimates, basis enlargement to $n_{\max} = 4$, an explicit-core Hartree–Fock treatment with twelve electrons, and a DMRG cross-check that reproduced FCI bit-identically at finite sector dimension without producing an interior minimum (Papers 19/20 [5, 6]; CHANGELOG v2.19.4–v3.87.0). The wall was localized to the Hamiltonian specification—the projection from the continuous Level-4 solver to second-quantized $(h_1, \text{eri}, e_{\text{core}})$ integrals—rather than to the treatment of correlation.

This paper answers the question that localization leaves open. The framework neglects the overlap between orbitals on different centers, treating a non-orthogonal basis as though it were orthonormal, and that neglect is what preserves the sparsity. Was there an alternative? Could one keep the genuine cross-center physics *and* the label-generated sparsity, by declining to transform the integral tensor at all?

The answer is no, and the reason is group-theoretic rather than technical.

A. A change of method, not a refinement

Relation to the continuous theory. Evaluating integrals over genuine two-center orbital products is, in itself, nothing new: it is standard Coulomb–Sturmian practice, developed over decades by Avery and co-workers on the many-center hyperspherical-harmonic foundation [14, 15], and the framework's cross-center one-body couplings are precisely the Shibuya–Wulfman integrals [16] of that theory (Paper 19 [5]). GeoVac adds nothing to that continuous apparatus. What is new is not the integrals but the *discrete* reading of them: which of the continuous theory's couplings survive as exact, label-forced zeros when the angular content is carried to a qubit Hamiltonian, and which do not. In that sense this paper is the discrete/combinatorial skeleton of a continuous subject—the relationship the angular selection rules already bear to their Gaunt and Wigner origins, now traced into the two-center case.

It is worth stating plainly what this paper does differently, because the step is easy to mistake for an incremental improvement in integral quality. It is not. Every previous GeoVac molecular builder evaluates its integrals in the *atomic sector*: each orbital product entering an integral sits on a single nucleus, and inter-nuclear coupling enters through a surrogate. Three generations, distinguished by what they do at the bond:

1. **Composed geometry** (Paper 17 [4]). A Phillips–Kleinman pseudopotential replaces explicit two-body core–valence Coulomb with one-body Z_{eff} screening, so *cross-block electron-repulsion integrals vanish identically.* The bond is not present in the two-body tensor at all.

2. **Balanced coupled composition** (Papers 19/20 [5, 6]). The pseudopotential is eliminated and cross-block repulsion is restored, balanced by a cross-center nuclear-attraction term. But both restored pieces remain atomic-sector objects: the cross-center one-body integral is $\langle\psi^A_{nlm}| - Z_B/|\mathbf{r} - \mathbf{R}_B||\psi^A_{n'l'm'}\rangle$, with *both* orbitals on center A and only the nucleus displaced, and the restored cross-block repulsion is constructed so as to continue obeying Gaunt triangle selection. The orbital products are still one-center; the l-selection rule still holds.

3. **This paper.** The integrals are evaluated over orbital products that genuinely span two nuclei, $\langle\chi^A| \cdot |\chi^B\rangle$. That is the step neither prior generation takes.

The distinction is not cosmetic, and the census below measures its price exactly. Because generations 1 and 2 never form a two-center orbital product, the Gaunt l-selection rule is exact for them by construction, and the label-generated sparsity follows. Once the product genuinely spans two centers, that rule fails—only the axial

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m-rule survives—and the permitted density inflates by 13.8× at $n_{\max} = 2$. The sparsity was never a discovered property of molecular electronic structure; it was a property of declining to evaluate the bond.

The demonstration in Sec. VI is the corresponding gain. NaH is the system the prior generations could not bind under any modification attempted, and it binds here with three fragment-native determinants. We therefore read W1e not as a defect to be repaired within the atomic-sector construction, but as the accumulated cost of that construction—which makes the present route a replacement for it rather than a correction to it. What the replacement forfeits is precisely the qubit-resource proposition Paper 14 rests on; Sec. V states that trade as a conservation law rather than a defeat.

Scope of this subsection. The three-generation description is a statement about what the builders compute, verifiable from their construction, not a new result. The demonstration below (Sec. VI) is a diatomic. A polyatomic extension has since been measured (BeH_2 and H_2O, basis ladder to $M = 19$): at matched minimal basis and matched correlation treatment the present route improves the geometry only modestly over generation 1 (+10.0% vs 11.7% for BeH_2, +14.9% vs 19.4% for H_2O), but the error is basis-limited rather than method-limited and falls to +1.4% and +2.5% respectively once the basis is enlarged—and the H_2O bond angle, an observable generation 1 cannot represent at all, comes out at 106.3° against an experimental 104.5°. Those figures are RHF over Gaussian-fitted Slater shapes in the same engine used for Sec. VI; they are reported here as scope, not as a result of this paper.

II. THE EXACT-ARITHMETIC CENSUS

A. Machinery and certificate

Two-center overlaps at arbitrary (n, l, m, Z) are computed as exact rationals via the Mulliken auxiliary functions A_n/B_n [8, 9]—one of several classical routes to two-center Slater integrals [20, 21]—evaluated in prolate spheroidal coordinates where the two-center problem separates [2]. Because the result is rational, vanishing is *decidable* rather than inferred from a numerical tolerance—a property no quadrature scheme can offer, and the reason this census can make a claim about exact zeros at all.

Cross-center one-body elements are obtained by extending the same route with $1/r_A$ and $1/r_B$ kernels, each of which cancels against the volume element $(\xi^2 - \eta^2)$, followed by the hydrogenic eigen-trick. The assembly carries its own certificate: the bra/ket identity

$$E_a S - Z_B I_B \;=\; E_b S - Z_A I_A \tag{1}$$

holds as an *exact rational equality* on a six-case panel and on every entry of the census configuration—and coefficient-wise on both exponential legs, which is strictly stronger than agreement at a single R. Its diagnostic power is itself tested rather than assumed: two independent corruptions of the inputs (a wrong bra eigenvalue, and I_A interchanged with I_B) each break the identity, so it is demonstrably not an algebraic tautology. Four independent quadrature cross-checks agree at $\sim 2 \times 10^{-16}$.

B. Result

The census configuration is $Z_A = 3$ at the origin, $Z_B = 1$ at $R = 3\,\hat{z}$, hydrogenic orbitals to $n_{\max} = 2$ on each center, $M = 10$ spatial orbitals. Table I compares the genuine bare tensor against the production builder's.

Provenance, stated precisely because the three rows differ, and because the rows are internally mixed. The cross-center blocks of S and h (22 entries each) are evaluated in exact rational arithmetic by the machinery of Sec. II A, so their *non*-vanishing is decided rather than inferred. The same-center S block is fixed analytically by orthonormality of same-Z hydrogenics rather than recomputed; the same-center h block is evaluated with the production float routine, so its zeros there are threshold decisions. And the vanishing of every non-matching-m entry is not evaluated at all—it is forced by Theorem 1(i). In this census the rational machinery decides non-vanishing; the zeros come from symmetry. The g row is *counted*: it is a symmetry-rule census in the complex-m basis, because no two-center electron-repulsion engine exists in the framework—the m-rule and per-side Gaunt feasibility are applied combinatorially, and no two-center electron-repulsion integral was numerically evaluated. The g figures are therefore claims about how many entries the selection rules *permit*, not verifications that each permitted entry is nonzero. Supplying that engine is the natural follow-on to this paper.

Every count in Table I is re-derived from the selection rules alone in `tests/test_paper58_census.py`. Both columns apply the *same* axial rule $m_p + m_r = m_q + m_s$; the builder column is the genuine column restricted to quartets whose four orbitals share one center, with the single-center shared-L Gaunt condition imposed identically in both (the same 176 removals at $n_{\max} = 2$). The inflation factor therefore measures the two-center span and nothing else—it is not an artifact of comparing a pair-diagonal m-convention against a global one, since

TABLE I. Genuine versus builder sparsity, $Z_A = 3/Z_B = 1$ at $R = 3\,a_0$, $n_{\max} = 2$ per center ($M = 10$). "Genuine zeros are" records which selection rule accounts for the surviving zeros.

object	genuine	builder	dense	genuine zeros are
S	32	10	100	m-rule + same-center
h	44	22	100	m-rule only
g	2,944	214	10,000	m-rule + same-center Gaunt

neither column uses the pair-diagonal rule.

Status of that follow-on. It is now partially supplied, and the per-class outcome bears directly on what the g row could become. Three of the four classes are closed in exact form and validated against independent engines: the one-center classes, $(AA|BB)$, and the hybrid pair $(AA|AB)/(AB|BB)$. For the exchange class $(AB|AB)$—Ruedenberg's own subject [10]—the assembly is validated for general (l, m) and its ordered radial double integral, the last non-closed step, now closes as well, for general (τ, σ, H, j) and unequal exponents (agreement with nested quadrature at 10^{-13}). All four classes therefore have closed-form machinery.

class	closed form	transcendental content
one-center, $(AA\|BB)$	yes	none—elementary
$(AA\|AB)$, $(AB\|BB)$	yes	$\{E_1, \ln\}$
$(AB\|AB)$	yes	$\{E_1, \ln, \gamma\}$

The transcendental content grows across the classes, but—and this is the structurally important part—it does *not* escalate in weight. The ordered radial integral is an iterated integral over a simplex, the shape that defines a period, and iterated integrals of weight-one objects generically land at weight two (dilogarithms, $\zeta(2)$). This one does not: assembled in closed form it agrees with quadrature to 6.3×10^{-16} and contains only exp, E_1, ln and Euler's γ. Nor does $\sigma \neq 0$ break it. The derivatives $d^\sigma Q_\tau/d\xi^\sigma$ carry poles of order up to σ at $\xi = \pm 1$, but they are absorbed exactly by the $(\xi^2 - 1)^H$ prefactor, because per electron the net exponent there is $(|m_a| + |m_b| - |m_a - m_b|)/2 \geq 0$ by the triangle inequality—with equality, i.e. exact absorption, whenever m_a and m_b have opposite signs. Verified to 2.5×10^{-14} with no weight-two content at any σ tested. So the seed sets above form a weight-one filtration throughout.

This does not upgrade the g row, for two independent reasons. First, no re-census has been run with the new machinery, so the counted figures stand exactly as counted. Second, and structurally, what the row needs is zero-*decidability*, and that is available only where a closed form exists.

For $(AA|BB)$ that argument now goes through, and by exactly the route the S and h rows already use. Every π cancels from the closed form— the harmonic normalisations against the $4\pi/(2L+1)$ of the multipole potential—leaving

$$(ab|cd) \;=\; A_0(R) \;+\; \sum_j A_j(R)\, e^{-\lambda_j R}, \qquad (2)$$

with A_j rational functions and λ_j rational. Checked on $(1s\,1s|1s\,1s)$, $(2p_0 2p_0|1s\,1s)$ and $(2p_{+1}2p_{+1}|2p_0 2p_0)$ across Z combinations: the π power is common (indeed zero) in every term. For algebraic R the exponents $-\lambda_j R$ are distinct algebraic numbers, so by Lindemann–Weierstrass the family $\{1, e^{-\lambda_j R}\}$ is linearly independent over the algebraic numbers and the expression vanishes *iff* every A_j does. That is the same separation the S and h rows are decided by, applied to a class that now has the closed form to support it.

The hybrid and exchange classes do not follow: they carry $\{E_1, \ln\}$ and $\{E_1, \ln, \gamma\}$ respectively, and would need an independence argument over those seeds, which we have not attempted.

Consequence for this paper's own sector. The classes carrying residual l-selection—$(AA|AA)$, $(BB|BB)$, $(AA|BB)$—are precisely the ones that are now zero-*decidable*: the one-centre classes trivially—they are exact rationals—and $(AA|BB)$ by the argument above. So the sector that Theorem 1 makes a positive prediction about is no longer merely counted but decidable.

And the decision has now been run on $(AA|BB)$. On the census configuration, that block holds 625 entries of which the counting rules permit 195. Deciding each one by the separation above—grouping the closed form by exponential rate and asking whether every rational coefficient vanishes—gives

outcome	count
decided nonzero	195
zero, vanishing Gaunt coefficient	0
zero, accidental (Gaunt nonzero, radial sum cancels)	0

All 195 permitted entries are genuinely nonzero. There are no accidental zeros and no symmetry zeros that the counting missed, so on this block the counted density is not an overcount but is *exactly* the true density. This is a decision rather than a measurement: no threshold is involved anywhere, and it upgrades the block from COUNTED to DECIDED. The same decision run over a 5×7 grid of charge pairs and internuclear distances—including the equal-charge and rate-coincidence configurations, where degenerate exponential rates are the likeliest source of an accidental cancellation, and a node-bearing $n_{\max} = 3$ sample—returns 195/195 nonzero at every point, so the verdict is not an artifact of the census configuration.

It also settles, for this block, the reading the Gaussian corroboration could only suggest. That check found 29.8% against 29.4% counted and attributed the gap to the real-Cartesian versus complex-m basis convention rather than to over-permissive counting; the decided census confirms there is no over-permissiveness here to find.

The g row's overall tier is nonetheless unchanged, because $(AA|BB)$ and the one-centre classes together are only the 20.5% that carries residual l-selection. The three genuinely cross classes—79.5% of the tensor—remain counted, and will until an independence argument over $\{E_1, \ln, \gamma\}$ is available.

We state the outlook carefully, because both natural readings are wrong. The pessimistic one—that the seed set escalates—is wrong: it is a weight-one filtration at every class, and a closed form now exists for every class. But the optimistic one—that the remaining independence argument is therefore *ordinary work*—is wrong too, because *weight one is not the same as decidable.* The $(AA|BB)$ decision went through only because that class is the one place π cancels *and* no E_1, ln or γ appears: pure $\{e^{-\lambda R}\}$, which Lindemann–Weierstrass separates at any algebraic R. The cross classes carry $E_1(\mu R)$ (hybrid and exchange) and, for the exchange class, a standalone Euler γ that survives assembly at every τ. Deciding an entry's vanishing at a fixed R then requires those seed *values* to be linearly independent—and E_1-value independence at algebraic points is an open problem of transcendence theory, while γ is not known even to be irrational. That is a genuine obstruction in principle, not ordinary work, and it draws a *categorical* line—$(AA|BB)$ decidable, the E_1/γ classes not—rather than the gradient the weight-one filtration alone suggests.

What *is* available now is one rung weaker and unconditional. The seed *functions* $\{1, e^{-\lambda R}, E_1(\mu R), \ln R\}$ are linearly independent over $\mathbb{Q}(R)$—**[SYMBOLIC (proof-by-argument)]** a theorem (Liouville–Rosenlicht: E_1 is the standard non-elementary integral, transcendental over the exponential field)—so every cross-class entry is nonzero *as a function of* R unless its grouped coefficients vanish, and any zero is therefore isolated at special R. With the Gaussian corroboration above already showing the permitted entries generically nonzero, the honest position for the cross classes is: functionally nonvanishing (unconditional), generically nonzero (measured), and pointwise-DECIDABLE only *conditional* on a Schanuel-type independence hypothesis for the E_1 and γ values. That is materially better than counted, and blocked from decided by open transcendence rather than by missing labour.

Beyond two centres (scope). The transcendence does not merely climb the weight ladder at the third centre; it changes ladder. The two-body three-centre integral $(XY|XZ)$—two two-centre densities sharing one centre on different axes, the first object a polyatomic requires and one no prolate spheroidal system can hold (that coordinate has exactly two foci)—is naturally evaluated in momentum space, where the Coulomb kernel is $4\pi/k^2$ and each centre is a phase rather than a focus. There the two densities Fock-project onto spheres of *different* radii $\sqrt{c_1} \neq \sqrt{c_2}$, and the product of their stereographic weights is the quartic $y^2 = (c_1k^2 + 1)(c_2k^2 + 1)$—an elliptic curve. Its $D = 0$ period is exactly a complete elliptic integral, and the full integral is a genus-one object on the same family—by Paper 59's final characterization an irreducible $\Gamma(2)$ period (the elementary elliptic-polylogarithm target is excluded there)—degenerating to the genus-zero two-centre case only on the measure-zero locus $c_1 = c_2$ of a shared sphere. The two negatives that make the two-centre engine weight-one and $(AA|BB)$ decidable—no π, no dilogarithm—are therefore *two-centre* properties: the third centre leaves the polylogarithm hierarchy for genus-one elliptic transcendentals. This is the transcendence-level form of the reason Gaussian orbitals, whose product theorem collapses three centres to one, displaced Slater orbitals for polyatomic quantum chemistry. It is reported here as scope; the method and the elliptic identification are numerical (momentum-space evaluation validated against an independent Gaussian reference to $\sim 10^{-14}$, and the elliptic period verified to 31 digits), not a closed form.

What the build does settle is narrower, and worth stating because it is exactly the sector this paper's mechanism concerns. The classes in which the residual l-selection survives—$(AA|AA)$, $(BB|BB)$ and $(AA|BB)$, together 20.5% of the genuine tensor—are now exactly computable in closed form and carry no transcendental seed whatever. An l-selection census restricted to those classes is therefore no longer gated on the missing engine. That is the sharpest thing the build supplies for this paper: the sector where Theorem 1 predicts residual l-selection to survive is now exactly computable, and computable with no transcendental seed at all, so a census over it would test the theorem's positive content directly rather than by counting.

The two halves compose. The census machinery of Sec. II supplies exact two-centre S and h—the Mulliken/Ruedenberg auxiliary route and the hydrogenic eigen-trick—and named the missing two-centre electron-repulsion engine as its one gap. With that engine supplied, the three ingredients now compose into a complete calculation carrying no Gaussian fit anywhere. Verified on H_2 at $R = 1.4$ bohr in a $1s$-per-centre basis: total FCI energy -1.106556606091 Ha. Evaluating the identical basis through Gaussian-fitted integrals converges monotonically onto that value as the fit deepens (2.7×10^{-3}, 3.0×10^{-4}, 4.5×10^{-5}, 8.0×10^{-6}, 1.9×10^{-6} Ha at $4, 6, 8, 10, 12$ primitives), confirming the native value is the exact one. The residual sits in h, not in g: the electron-repulsion tensors agree to 1.9×10^{-8} and the overlaps to 1.5×10^{-8} at ten primitives, while the one-electron terms lag because kinetic energy probes the nuclear cusp, where a sum of Gaussians converges slowly by construction.

Subject to that distinction, the electron-repulsion tensor is 29.4% permitted-dense genuinely against 2.1% for the builder.

Numerical corroboration of the g row. The counting can be checked without the missing Neumann engine [10], by evaluating the same tensor with a McMurchie–Davidson Gaussian engine [12] over fitted Slater shapes. On the census configuration this gives 2,976 of 10,000 entries above 10^{-10}, i.e. 29.8% dense against the 29.4% counted—agreement at the level the basis convention permits, since the numerical evaluation is in the real Cartesian basis and the count is in the complex-m basis, which differ by the $\pm m$ mixing. So the permitted entries are *generically nonzero*: the density is a real property of the tensor and not an artifact of over-permissive counting. The l-selection failure appears at bond scale in individual entries—for instance $(1s_A\, 2p_{z,B}\,|\,2s_A\, 2s_A) = -0.218$, where the same-center analog vanishes identically.

This does not upgrade the row to MEASURED: Gaussian-fitted floats make these threshold decisions rather than decided zeros, so the exact statement still awaits the engine. What it removes is the possibility that the counted density was an overcount.

The m-rule, verified without reference to m. Theorem 1(i) has a basis-free consequence that can be tested directly on the tensor: since $1/r_{12}$ is rotationally invariant and both nuclei lie on the axis, a rotation about the internuclear axis maps the basis into itself and must leave the tensor fixed. It does, to 2.2×10^{-16}. Displacing the nuclei off that axis and repeating the identical check gives 1.3×10^{-1}—fifteen orders of magnitude larger. The symmetry is axial, and only axial. At $n_{\max} = 3$ ($M = 28$) the comparison is 114,280 *permitted* entries (18.6%) against 7,600 (1.2%): the inflation factor is 13.8× at $n_{\max} = 2$ and 15.0× at $n_{\max} = 3$, i.e. *increasing* with basis size. Both are counted; only the $n_{\max} = 2$ figure is numerically corroborated below.

Two further measurements matter for interpretation. First, per-side Gaunt feasibility removes 5.6% of entries at $n_{\max} = 2$ and 6.3% at $n_{\max} = 3$, beyond what the m-rule already removes, and the three cross classes $(AA|AB) + (AB|AB) + (AB|BB)$ account for 79.5% of the genuine tensor at $n_{\max} = 2$ (80.0% at $n_{\max} = 3$). The residual l-selection is therefore confined to the classes carrying a same-center distribution on each side—$(AA|AA)$, $(BB|BB)$ and $(AA|BB)$—together 20.5% of the genuine tensor; the three genuinely cross classes carry none. Second, cross-center magnitudes reach bond scale: the largest overlap is 0.52 and the largest one-body entry is 0.80 Ha. These are *maxima*, not a range—the census block also contains much smaller entries (the smallest overlap is 0.011, on the $(2p_0, 2p_0)$ pair). An earlier statement of them as intervals 0.08–0.52 and 0.19–0.80 took its lower ends from an inconsistent subset of the census pairs and was incorrect. For scale, the entire NaH bond is under 2 eV, so individual entries in the dense part of the tensor exceed the binding energy being computed. They are not droppable as small corrections—an independent quantification of the W1d/W1e diagnosis.

III. MECHANISM: THE ABELIAN RESIDUE

The census result is not an accident of the chosen configuration. It is forced, and the forcing is elementary once stated in the right language.

Theorem 1 (Angular selection survives molecularization only in the abelian sector)**.** *Let two nuclei A, B lie on the z-axis, with atom-centered basis functions labeled (n, l, m) about their own center, and let $\hat{O}$ be invariant under the molecular symmetry group. Then:*

(i) The m-selection rule holds exactly across centers: $\langle ab|\hat{O}|cd\rangle = 0$ unless $m_a + m_b = m_c + m_d$.

(ii) No l-selection rule holds between basis functions on different centers. More strongly, the cross-center support is full *in l at fixed m: the re-expansion does not terminate, so every admissible l couples.*

(iii) Of the per-center $SO(3)$ labels, only those of the common axial subgroup $U(1) \cong SO(2)$ survive cross-center; by Pontryagin duality its characters are the integers, so the surviving cross-center *rule is additive m-conservation.*

Part (iii) is deliberately narrow, and two limits are worth stating because the wider reading is false. It does not say the axial $U(1)$ generates all surviving sparsity: same-center Gaunt selection survives from the full per-center $SO(3)$ (visible in Table I's g and S rows), and the largest group fixing both nuclei is not $U(1)$ but $C_{\infty v}$, which is *non-abelian*—its σ_v reflection is exactly the Hopf $m_l \to -m_l$ $\mathbb{Z}_2$ used as the baseline of Table II. So a non-abelian common group does yield usable label structure. What (iii) claims is only that the *continuous* per-center angular labels reduce to the axial one across centers, which is what the census measures.

Proof sketch. For (i): rotation about the internuclear axis is a *single* one-parameter group acting on all of $\mathbb{R}^3$. Its generator L_z is independent of which point on the axis is taken as origin, so m is a global label shared by both centers rather than a per-center one. Invariance of $\hat{O}$ under this group gives additive conservation of m.

For (ii): l labels irreducible representations of the rotation group about a *specific* center. Rotations about A and rotations about B are distinct subgroups of the Euclidean group whose intersection is precisely the axial subgroup. A function that is an l-eigenfunction about B, re-expanded about A, has support on all l: the two-center re-expansion does not terminate. Hence no l-selection between cross-center pairs. (The same-center blocks retain Gaunt selection, consistent with the 5.6%–6.3% residual measured in Sec. II.)

For (iii): the intersection in (ii) is $SO(2) \cong U(1)$, abelian. Abelian compact groups have one-dimensional irreducibles labeled by characters, and by Pontryagin duality the character group of $U(1)$ is $\mathbb{Z}$—which is the integer label m. Selection rules from abelian symmetry are additive conservation laws on integer charges, and are preserved by any operation respecting the axis. Non-abelian selection rules are Clebsch–Gordan couplings requiring the full group; reducing the group destroys them. □

a. Prior art: what is standard here and what is not. The ingredients of Theorem 1 are classical, and we claim none of them. Step (ii)'s intersection rule—the residual symmetry of a system under an external influence is the largest subgroup common to the system and the influence—is *Curie's principle*, stated in 1894 [38] and standard in crystal physics in the form $\tilde{K} = K \cap G$. Step (i)'s reading of surviving labels as irreducible representations of the residual subgroup is descent in symmetry in the sense of Bethe [39], the foundation of crystal- and ligand-field theory. Step (iii)'s identification of the characters of a compact abelian group with a discrete additive charge is elementary Pontryagin duality. What we have not found stated as a general principle is the abelian/non-abelian *asymmetry* itself—that additive selection survives subgroup restriction while Clebsch–Gordan selection does not—which appears in the literature as case-specific folklore (Λ in diatomics, m_j in the Paschen–Back regime, k in solids) and as an engineering constraint in symmetry-adapted quantum algorithms. We state it here for completeness rather than for priority.

What is ours is the *target*. The prior literature applies symmetry descent to predict spectroscopic level splittings; the use made of it here is to predict the *sparsity of the two-center integral tensor*—which l-selection dies, and by how much the symmetry-permitted density inflates (Sec. II)—which is a statement about the cost of a molecular Hamiltonian, not about a spectrum.

b. The correlation-diagram reading. **[OBSERVATION]** Theorem 1 is the group-theoretic content of a construction every chemist already uses. The Mulliken correlation diagram of a diatomic [23, 40, 41] connects the united-atom labels at $R \to 0$—where the prolate-spheroidal η-equation's separation parameter reduces to l (Paper 11 [2])—to the separated-atom labels at $R \to \infty$, where each center carries its own l. The one rule that organizes the connections is the non-crossing rule of von Neumann and Wigner [22]: as R varies, two levels of the same symmetry cannot cross. For a linear molecule that symmetry is exactly the surviving axial character m (the $\sigma/\pi/\delta$ label), together with g/u for a homonuclear pair and, for many-electron states, spin. Read along R, part (i)—m conserved cross-center at every R—supplies the classification the non-crossing rule sorts by, and part (ii)—l not shared—is why there is a diagram to draw at all: were l shared, united-atom and separated-atom labels would coincide and the diagram would be trivial. So the sparsity that survives molecularization and the $\sigma/\pi/\delta$ bookkeeping that governs orbital crossings are the same axial $U(1)$ seen from two sides. This is an identification with a standard result, not a new claim; the theorem's backing is unchanged.

The theorem is independently corroborated by the framework's own production code, along a path that knows nothing of this analysis: the cross-center nuclear-attraction routine in `shibuya_wulfman` returns zero by an explicit early exit when $m_1 \neq m_2$, and imposes no l-selection on the element: its per-multipole Gaunt conditions (triangle inequality and $l_1 + L + l_2$ parity) are satisfiable by some L for every (l_1, l_2), so no l pair is excluded. The m-rule is hard-coded because it holds; the l-rule is absent because it does not.

Theorem 1 converts a measurement into a prediction, because the largest common abelian symmetry is readable from the molecular point group before any integral is computed.

Corollary 1 (Sparsity budget from the point group)**.** *For a molecule with point group G, label-generated angular sparsity is bounded by the grading available from the abelian structure of G. Linear molecules ($C_{\infty v}$, $D_{\infty h}$) carry a continuous $U(1)$ and therefore a $\mathbb{Z}$ grading, worth a multiplicative factor. Bent molecules carry only a finite point group and therefore a finite grading.*

H_2O has point group C_{2v}, which is order four and *abelian*, isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2$; all its irreducibles are one-dimensional, so Corollary 1 predicts a two-bit spatial grading—two qubits, not a multiplicative factor. We attempted to test this and could not, for a reason that is itself a structural finding.

Observation 1 (The composed builder carries no angular geometry)**.** *`molecular_spec.py` contains no bond angle: `h2o_spec` is `hydride_spec(8)`, parameterized by nuclear charge and a* single *O–H distance distributed across five blocks (core, two bond pairs, two lone pairs), with `MolecularSpec.nuclei = None`. The composed* H_2O *Hamiltonian therefore does not represent C_{2v} at all, and the framework's reported* H_2O *geometry error is a purely radial coordinate with no angular degree of freedom. Corollary 1 is consequently* not well posed *on this builder in the bent case; supplying hand-built bent nuclei would test a symmetry the Hamiltonian does not encode.*

Prediction 1 (Conditional, for any angular build)**.** *In a build that does carry angular geometry, a molecule whose point group is abelian of order 2^k admits a k-bit spatial grading and no more. This remains falsifiable, but not on the present builder.*

What *is* measurable on the present builder is the linear case, where geometry is angle-free and therefore unambiguous—and it returns a result we did not anticipate, reported as Observation 3 below.

The measured factor of 3–5 for the m-rule at the bases of Sec. II is the $\mathbb{Z}$-grading case of Corollary 1. It is exact and structural, and it is a constant factor—not the exactly-linear-in-Q Pauli-count story ($N_{\text{Pauli}} = 27.90 \times Q$ across molecules at fixed basis, Paper 14) the framework's qubit results rest on.

IV. THE CONTINUOUS SIDE: THE DECOMPACTIFICATION FRONT

Theorem 1 is a statement about *labels*, and labels are all-or-nothing: m is shared at every $R > 0$ and l at none. That leaves open what a chemist actually watches as two atoms approach—how the coupling that l used to forbid turns on. (As a symmetry statement this is the continuous breaking of the united-atom $O(4)$ degeneracy; the continuous adiabatic terms of the two-center Coulomb problem are the subject of Solov'ev's hidden-crossings program [31, 32], which connects the unitedand separated-atom limits through that same $O(4)$ symmetry. What the front below adds to that program is a *metric* for the transition—a projector-commutator criterion and a measured decay-length law—not the qualitative fact of a continuous connection.) This section measures that—first for the one-electron case, then its many-electron extension—and finds it continuous in R with a length scale the reader can name. (The many-electron front, and the one place correlation enters, are developed below; the complementary statement that the Shibuya–Wulfman metric does not *compound* with electron number is in Paper 60 [24], Sec. "the metric does not compound".)

a. The per-shell front. **[MEASURED]** Take hydrogenic ns orbitals of charge Z on both centers and let $S_n(R) = \langle ns_A|ns_B\rangle$. The two single-center projectors onto these shells fail to commute by $\|[P_A, P_B]\| = S_n\sqrt{1-S_n^2}$, maximal ($=\frac{1}{2}$, principal angle 45°) at $S_n = 1/\sqrt{2}$ [29, 30]. Define the *front* $R^*(n)$ as the distance at which that maximum is reached. It tracks the orbital's exponential decay length $\ell_n = n/Z$, not its mean radius $r_n = n^2/Z$:

n	R^*	R^*/ℓ_n	R^*/r_n
1	1.565	1.565	1.565
2	4.39	2.19	1.10
3	6.57	2.19	0.73
4	8.52	2.13	0.53
6	12.82	2.14	0.36
8	17.15	2.14	0.27

($Z = 1$; bohr). For $n \geq 2$ the decay-length ratio lies in 2.12–2.19 (median 2.14, flat within 3%) while the mean-radius ratio falls by 4×; the fitted exponent is $R^*(n) \propto n^{0.98}$, i.e. a front *linear* in R, $n^*(R) \simeq ZR/2.14$. The reason is that overlap across a gap is a tail phenomenon: it is governed by how far the exponential tail reaches, not by where the bulk of the density sits. *The criterion itself is inherited, not new here*: that the large-separation coupling of two atoms is set by the exponential decay rate of the tails rather than by any mean radius is the content of Herring's critique of the Heitler–London method [42] and of the asymptotic exchange coupling of Herring and Flicker [43]; the same criterion recurs as the e^{-mL} control of finite-volume corrections in lattice field theory. What is measured here is the constant and the exponent for a *projector-commutator* front, $\|[P_A, P_B]\|$, where we have not found the criterion applied. The naive criterion "$r_n \approx R$" (a window at $n \sim \sqrt{ZR}$) predicts exponent $\frac{1}{2}$ and is rejected by the same data. Backing: `tests/test_paper58_decompactification_front.py`, which recomputes every entry by an independent prolate-spheroidal quadrature route and also rejects the mean-radius scaling explicitly. The companion curves—the bond-sphere angle $\gamma(R)$ of Paper 8 [25], the principal angles [26–28] between the two center projectors (which reproduce the LiH values 7.6°/44.7°/67.3° of Paper 32's composition remark at its bond length), the aggregate l-mixing, and the argument of the closed-form seed $e^a E_1(a)$ of Sec. VII, $a(R) = R$—are all smooth in R (chronicle: CHANGELOG v5.10.2). **[OBSERVATION]** So the picture is: the label is Boolean, the coupling is continuous, and the continuous knob is the decay length. The abelian residue is the Boolean half of the two-center problem; this front is its continuous half; the correlation-diagram reading of Sec. III is where the two meet.

b. The continuity is analytic, with a bounded rate. **[SYMBOLIC + MEASURED]** The smoothness asserted above is not only observed; for the angular label it is a known property of the special function involved, and this fixes its rate. Write the $m = 0$, homonuclear η-equation of Paper 11 [2] as $\frac{d}{d\eta}\left[(1-\eta^2)G'\right] + (-A + c^2\eta^2)G = 0$ and compare the spheroidal equation in the form of DLMF §30.2 [44]. Matching term by term gives the dictionary

$$\gamma^2 = -c^2, \qquad \lambda_n^m(\gamma^2) = c^2 - A, \tag{3}$$

so the framework's angular separation constant *is* a spheroidal eigenvalue in disguise. Three consequences follow, all verified against the solver:

- DLMF (30.3.3), $\lambda_n^m(0) = n(n+1)$, becomes $A(c^2 = 0) = -n(n+1)$: reproduced to twelve decimal places for every (m, n) with $m \leq 2$, $n \leq 4$. This is the united-atom limit in which the separation constant degenerates to the Legendre label—the $R \to 0$ end of the correlation diagram, now pinned to a special-function identity rather than asserted.

- DLMF §30.3(ii), analyticity of λ_n^m in the real variable γ^2, upgrades "continuous in R" to *analytic* in c^2. The Boolean/continuous split of this section is therefore sharper than measurement alone establishes: the label is discontinuous at $R = 0$ while the coupling is real-analytic throughout.

- DLMF (30.3.4), $-1 < d\lambda_n^m/d\gamma^2 < 0$, becomes

  $$0 < \frac{dA}{d(c^2)} < 1 \tag{4}$$

  under Eq. (3). Measured over nine modes at forty values of $c^2 \in [0.25, 60]$, the slope lies in $[0.14, 0.87]$ with no violation. The decompactification of l therefore proceeds at a rate that is strictly positive and strictly bounded above—it cannot stall, and it cannot outrun c^2.

- DLMF (30.3.1), $\lambda_m^m < \lambda_{m+1}^m < \cdots$, holds at every c^2 tested, and following it to large c^2 reaches the *other* end of the correlation diagram. The levels collapse into near-degenerate doublets: at $c^2 = 140$ the within-doublet gap is $\sim 10^{-7}$ while the between-doublet gap is $O(10)$, a ratio above 10^6—the σ_g/σ_u pairing of the separated-atom limit. **[MEASURED]** The gap closes as $\sim c^2 e^{-2c}$: the local slope $d\ln(\mathrm{gap})/dc$ rises monotonically from -1.69 to -1.89 over $c \in [6.3, 17.3]$, approaching -2 from above, and dividing out e^{-2c} leaves a power law $c^{2.07}$ at $R^2 = 0.999$. (A single exponential fitted naively to the same data returns -1.67; that is the algebraic prefactor contaminating the slope, not a decay rate, and it is the reason the two constraints are asserted separately in the backing test.) Since $c \propto R$ at fixed energy scale, this is an exponential in the internuclear distance—the exchange-splitting class of Herring [42, 43], the same asymptotics cited above for the front. *The two ends of the diagram are thus both pinned:* DLMF (30.3.3) fixes the united-atom limit to the Legendre label, and this doubling fixes the separated-atom limit; what happens between them is the analytic, rate-bounded interpolation of the two preceding items. We report the splitting law as measured on this branch and do not attribute it to a literature asymptotic: the quantity splitting here is the separation constant λ, not an energy, and the map from $\Delta\lambda$ to the physical σ_g/σ_u energy splitting runs through the radial equation.

Two honesty notes. DLMF states (30.3.4) for the real variable γ^2 without an explicit domain restriction, and the branch this problem occupies is $\gamma^2 = -c^2 < 0$ (the oblate branch); we therefore report the bound as *verified numerically* on that branch rather than cited as proved there. Second, the check doubles as an independent-route validation of the angular solver itself: its eigenvalues agree with `scipy`'s oblate spheroidal characteristic values at the 10^{-13} level across the grid, a correspondence the module docstring asserted but which was not previously tested. Backing: `tests/test_paper58_dlmf_spheroidal.py`; driver `debug/dlmf_spheroidal_crosscheck.py`.

c. Two unequal tails. **[MEASURED]** For $1s$ tails of different decay lengths ℓ_A, ℓ_B (charges $Z_A \neq Z_B$), the tail-reach front R_{rel}—the distance at which $|S|$ falls to $|S|_{\max}/\sqrt{2}$— combines the two lengths as a *geometric mean* [33],

$$R_{\mathrm{rel}} \simeq 1.58\sqrt{\ell_A \ell_B}, \tag{5}$$

constant to 2% over the charge pairs $(1,1), (2,1), (3,1), (2,2)$ and to 9% over a dense exponent-ratio scan $t = \zeta_B/\zeta_A \in [1, 8]$, where the additive law $c(\ell_A + \ell_B)$ misses by 24%, the longer-tail law by 62%. **[SYMBOLIC]** The *absolute* 45° front, by contrast, exists only when the united-atom overlap $S(0) = \left(2\sqrt{t}/(1+t)\right)^3$ exceeds $1/\sqrt{2}$, i.e. for $t < t_c = 2.664$; beyond that ratio the two shapes are already more than 45° apart at $R = 0$ and, $S(R)$ being monotone decreasing for every t, no distance brings them closer. Mixed-n pairs are near-orthogonal at every R and carry no front. Backing: the same test file (`test_geometric_mean_tail_law_1s1s`, `test_absolute_front_threshold_t_c`). Both facts are inputs to the many-electron front developed next, whose prediction uses the exact $1s$–$1s$ overlap at the measured per-center exponents rather than either fitted constant.

d. The many-electron front. **[MEASURED]** The front extends to many electrons through the compound-matrix identity: a k-electron configuration overlap is the k-th compound of the one-electron overlap [17, 27, 36, 37], so many-body physics can move the one-electron front R^* above *only* by changing the decay length the electrons actually occupy. We tested this on H_2 and HeH^+ at Hartree–Fock and full CI in a per-center multi-ζ $1s$ basis (five, ten, and eight exponents per center; the effective per-center exponent $\zeta_{\mathrm{eff}}(R)$ fitted from the one-body density's own tail, and verified to be an *output*: it runs $1.46 \to 0.83$ across R at HF, reaches the textbook 1.19 at $R = 1.4$, differs between HF and FCI by 12% at $R = 4$, equals no basis exponent, and moves the front by $\leq 0.8\%$ under basis doubling). The occupation-weighted principal-angle front of the occupied one-body space matches the exact $1s$–$1s$ law at the empirical ζ_{eff} to -0.7% (H_2, HF), -1.2% (H_2, FCI), 0.0% (HeH^+, HF) and -0.2% (HeH^+, FCI), with $|\Delta\cos| \leq 0.006$ pointwise wherever the second center holds more than 1% of an electron. **Screening moves the front; nothing else does, at the one-percent level.** The sub-percent residual is a shape term (a contracted multi-ζ component is not one exponential), non-positive at every rung.

[SYMBOLIC + MEASURED] Correlation does move a *different* front, and through occupation numbers rather than any length. Let γ be the one-body density and P_{XY} its center-blocks; the *signed* cross-center coherence $M_2 = \mathrm{Tr}(P_{AB}S_{BA})/\sqrt{\mathrm{Tr}(P_{AA}S_{AA})\,\mathrm{Tr}(P_{BB}S_{BB})}$ coincides with the principal-angle front at HF (one occupied orbital) but crosses $1/\sqrt{2}$ inward at FCI by 23% (H_2: 0.987 vs 1.285 bohr) and 9% (HeH^+: 0.690 vs 0.758). In the two-orbital form the mechanism is explicit: with natural-orbital occupations n_g, n_u and per-center weights $w_g^2 = 1/(2(1+S))$, $w_u^2 = 1/(2(1-S))$,

$$M_2 = \frac{n_g w_g^2 - n_u w_u^2}{n_g w_g^2 + n_u w_u^2}\,S, \qquad \frac{w_u^2}{w_g^2} = \frac{1+S}{1-S}, \tag{6}$$

so the antibonding orbital enters the per-center density with a weight amplified about fivefold at $S = 0.67$, and $n_u = 0.025$ already pulls M_2 from $S = 0.674$ to 0.591 (measured 0.596). The limits are the molecular-orbital value $M_2 = S$ ($n_u = 0$) and the Heitler–London value $M_2 = S^2$ (at HL occupations $n_g = (1+S)^2/(1+S^2)$, $n_u = (1-S)^2/(1+S^2)$); H_2 runs from the first toward the second as R grows. This is bond-order collapse—the k-th compound of the same one-electron overlap—so correlation touches decompactification in exactly one place, and that place is occupation, not geometry. Scope: s-only per-center basis, two electrons; whether the principal-angle statement survives at $N > 2$ with several strongly occupied natural orbitals is untested. Backing: `tests/test_paper58_coherence_front.py` (the limits of Eq. (6) symbolically, the measured H_2 point, and the rejected equal-weight alternative); chronicle CHANGELOG v5.10.2.

V. TWO RE-READINGS

A. Löwdin was paying an honest bill

The corpus previously recorded a 17.9× Pauli inflation from retrofitting Löwdin orthogonalization [11] onto the balanced LiH integrals (CHANGELOG v4.73.1), and a 14× inflation in the earlier heterogeneous-nested experiment. Both were read as the price of orthogonalization, and orthogonalization was accordingly treated as the sparsity-destroying move.

The census shows this reading was wrong in an instructive way. The genuine tensor's own permitted-density inflation over the builder is $\sim 15\times$—the same order as the 17.9× Löwdin figure, though the two are not commensurable: 17.9× is a Pauli-term count on Löwdin-retrofitted balanced LiH integrals, 15.0× a counted permitted-entry ratio on a $Z_A = 3/Z_B = 1$ hydrogenic census, and neither carries an error bar. The inference does not need them to agree numerically: the *bulk* of that inflation cannot have been orthogonalization's price, because the untransformed tensor never possessed the sparsity Löwdin was said to destroy. "Löwdin destroys sparsity" and "the molecular tensor is intrinsically l-dense cross-center" are the same fact observed twice, from two sides.

This matters because it removes an entire class of proposed remedies. Any scheme whose premise is "avoid the transformation and keep the sparsity" is answering a question that was mis-posed.

B. Cost conservation

Observation 2 (Relocating the metric does not reduce it)**.** *Five independent attempts to move the non-orthogonality out of the way each succeeded in moving it and failed to reduce it:*

1. *into the **operator** (Löwdin $S^{-1/2}$): operator densifies* 17.9×*;*
2. *into a **dual basis** (biorthogonal encoding): densifies comparably. The metric provably does not disappear—in the contravariant representation it reappears as an S^{-1} contraction, and the dual creation/annihilation operators cease to be Hermitian conjugates [13]—but whether that S^{-1} is* dense *is a property of the two-center Slater basis, measured here rather than imported;*
3. *into a **high-symmetry embedding** (hyperspherical Level 4N): angular basis requirement explodes—LiH at* 63.5% R_{eq} *error with $l_{\max} = 2$ genuinely insufficient;*
4. *into **device-measured many-body overlaps** (NOCI): succeeds at the state level, but the operator is dense independently;*
5. *into a **permutation symmetry** (equivalent-atom swap): net loss, measured—see Observation 3.*

The invariant is that the cost measures the physical asymmetry of a two-center problem. The currency is a choice; the total is not.

The fifth entry was measured for this paper and is the sharpest instance, because the trade is visible term by term.

Observation 3 (Permutation symmetry costs qubits on this builder)**.** *Enabling equivalent-atom-swap tapering* reduces *the qubit saving and inflates the Pauli count, in every swap-eligible configuration tested (Table II). The mechanism is visible in the retained stabilizer labels: the swap rotation merges the per-block Hopf and ℓ-parity stabilizers into joint `SWAP_JOINT` stabilizers, so one permutation stabilizer is bought at the price of several per-block abelian gradings. **[MEASURED]***

This is consistent with Theorem 1 rather than incidental to it. A permutation is not an abelian grading of the existing block structure; it is an operation that *identifies* blocks, and identifying blocks destroys the per-block labels that generated the sparsity. The abelian residue is not merely what survives—it is what can be exploited without collapsing the structure that carries it.

Documentation discrepancy, since corrected. The docstring of `extended_tapered_from_spec` previously stated that atom-swap "saves additional qubits on polyatomics (BeH_2 +1, H_2O +1, NH_3 +1, CH_4 +2)." The measured BeH_2 value is −1 against the Hopf-only baseline and −3 against Hopf + ℓ; the sign is wrong under both. We report the measurement and do not attempt to adjudicate whether the documentation was always incorrect or the behavior regressed.

Observation 2 is labeled an observation, not a theorem: it is a pattern across five instances with a plausible cause, not a proof that no sixth currency exists. The underlying obstruction has been characterized in the framework as non-commuting center projections, with

TABLE II. Equivalent-atom-swap tapering against two baselines. ΔQ is qubits removed (larger is better); Pauli counts are for the tapered operator. LiH has no equivalent atoms and is the null control.

system	baseline	ΔQ	ΔQ +swap	Pauli ratio
BeH_2	Hopf only	7	6	3.50×
BeH_2	Hopf + ℓ	12	9	3.60×
N_2	Hopf + ℓ	22	19	2.30×
F_2	Hopf only	12	9	3.50×
F_2	Hopf + ℓ	22	15	3.60×
LiH	Hopf + ℓ	8	8	1.00×

$\|[P_A, P_B]\| = 0.50$ at LiH equilibrium and principal angles $7.6°/44.7°/67.3°$.

VI. DEMONSTRATION: NAH BINDS WITH GENUINE INTEGRALS

If the wall is the Hamiltonian specification, then supplying genuine end-to-end integrals should bind NaH *without needing more* correlation than the failing runs used. It does—with considerably less.

Two scope conditions govern what follows, and the argument is weaker without them. First, this construction **makes no sparsity claim**: the radial functions are six-Gaussian fits of Slater shapes evaluated with a McMurchie–Davidson engine [12] and a dense non-orthogonal Slater–Condon solver, so it forfeits precisely the label-generated sparsity this paper is about—it is item 4 of Observation 2, not an escape from it. Second, the correlation treatment is not held fixed: the framework's failing runs were FCI and DMRG in an orthogonal second-quantized space, whereas this is three non-orthogonal determinants. That asymmetry is what makes the inference go through—binding appears with *less* correlation, so correlation cannot have been what was missing.

The construction is all-electron 12-electron NaH with Na$\{1s, 2s, 2p\times 3, 3s\}$ and H$\{1s\}$ ($M = 7$ spatial orbitals), non-orthogonal determinants built from fragment-native orbitals [18, 19], and general-N non-orthogonal Slater–Condon evaluation by Löwdin cofactors [17]. Orbital exponents were optimized on the *isolated* Na atom only ($E(\mathrm{Na}) = -161.0845$ Ha against a Clementi minimal-STO reference of ≈ -161.12), with hydrogen at $\zeta = 1$, so the molecular curve is a fragment prediction rather than a fit. Machinery validation at $R = 3.5\,a_0$: rotational invariance at 1.1×10^{-13}, and the 91-determinant non-orthogonal complete space reproduces the Löwdin bitstring FCI to 1.8×10^{-15} Ha. (The value serialized by the exploratory driver was corrupted by variable shadowing—the record was written after a geometry sweep that rebinds the nuclear-repulsion term—and has been corrected; the offset was exactly $11/3.5 - 11/10 = 143/70$.)

Three determinants give an interior minimum at +4.8% of the experimental bond length, with the in-basis FCI landing at +0.8%. The D_e shortfall—55% of experiment retained, so a 45% deficit—is minimal-basis incompleteness (no polarization, no diffuse functions, and no basis-set-superposition correction, the atom references being computed in atom-only bases) and reproduces the pattern of the four-electron LiH case at 53%; it is not an integral-fidelity effect.

As with LiH, the reading is sharper at *fixed* geometry, where no dissociation scan enters and the numbers are pinned by test. At $R = 3.5\,a_0$, against an isolated-atom dissociation reference of -161.584337 Ha and an in-basis FCI binding of 1.170 eV: two covalent determinants recover 50.9% of that binding, three recover 89.3%, and all four 90.6%—while the *same* three on Löwdin-orthogonalized orbitals recover 10.8%, and the Löwdin two-determinant case is not merely unbound but actively repulsive relative to the separated atoms (−153%). The well-minimum figures in Table III and these fixed-geometry figures tell the same story by two routes.

The Löwdin comparators are the informative rows. Orthogonalizing the orbitals and truncating to the *same* number of determinants retains 11.6% of the binding, and the two-determinant case is destroyed outright. Because the complete space is bit-identically the same either way, the damage is specific to *truncated* spaces. The four-electron LiH case replicates this at 15.0% and destroyed, respectively.

Observation 4 (Orthogonalization damages only truncated spaces)**.** *Orthogonalization is span-preserving and therefore harmless on the complete space, and severely harmful on truncated subspaces built from physically meaningful fragment structures. A qubit register only ever represents a truncated space.*

Observation 4 holds at four and twelve electrons, first and second row, with and without p orbitals—the four-electron leg pinned by test, the twelve-electron leg on the exploratory driver. We expect it to generalize, and it is cheap to falsify.

The mechanism is also visible at *fixed* geometry, without reference to a dissociation limit, which makes it the cleanest form to check. On LiH at $R = 3.25\,a_0$ in a three-orbital fitted-Slater basis, against a complete-space FCI of -8.887049 Ha: three fragment-native determinants land +1.32 mHa above it, while the *same three* determinants built on Löwdin-orthogonalized orbitals land +40.52 mHa above—a 30.6× larger error at identical determinant count. At two determinants the errors are +19.4 mHa and +118.7 mHa (6.1×). The complete spaces agree to machine precision, so the damage is at-

TABLE III. NaH, all-electron 12e, $M = 7$. Percentages are of in-basis FCI binding. Experiment: $R_{\mathrm{eq}} = 3.5667\,a_0$ ($r_e = 1.8874$ Å) [7]. The comparison value $D_e = 1.961$ eV is the primary spectroscopic determination of Huang *et al.* [34] (15815 ± 5 cm^{-1}; corroborated by the NaD isotope-shift analysis, and refined to 1.960 eV by Chu *et al.* [35]). Huber–Herzberg/NIST predate this measurement and carry no confident D_e, so the value is cited to its source rather than to the r_e compilation. The tabulated R_{eq} are the driver's coarse-grid ($0.25\,a_0$) parabolic minima; a finer grid shifts the 3-det value to $3.72\,a_0$, negligible within the minimal-basis error.

method	R_{eq} (a_0)	D_e (eV)	% FCI
covalent (2 det)	3.954	0.686	58.4
$+\mathrm{Na}^+\mathrm{H}^-$ (3 det)	**3.736**	**1.071**	**91.1**
all 4 det	3.713	1.080	91.9
FCI (91 det)	3.595	1.175	100
Löwdin covalent (2 det)	—	unbound	−38.0
Löwdin 3 det	3.658	0.136	11.6

tributable to truncation and not to the orthogonalization corrupting the integrals.

VII. THE QUADRATURE-FREE DIATOMIC

[**MEASURED + SYMBOLIC**] The engine's classes have now been used *end-to-end*: complete diatomic FCI energies in which every one- and two-electron integral is a closed-form expression—rationals, $\sqrt{\ }$, π, exp, E_1, log, γ_E (the π's are angular-measure and normalization prefactors of the *assembled* integrals; the radial seed content of each class is the $\{\exp, E_1, \ln, \gamma_E\}$ set of the preceding sections, and (AA|BB) is where even the prefactor π cancels)—evaluated to arbitrary precision, with no quadrature routine anywhere in the production path (`geovac/qfd_core.py`, `geovac/qfd_assemble.py`; `tests/test_paper58_qfd.py`).

Two pieces had to be added. *First*, the one-electron two-center closed forms (overlap, kinetic, both nuclear-attraction kernels) did not previously exist in the engine; they are derived from the Mulliken auxiliary integrals $A_m(p) = \int_1^\infty \xi^m e^{-p\xi} d\xi$, $B_n(q) = \int_{-1}^1 \eta^n e^{-q\eta} d\eta$ in prolate spheroidal coordinates, where the Jacobian factorization $(\xi^2 - \eta^2) = (\xi + \eta)(\xi - \eta)$ renders both $1/r_A$ and $1/r_B$ kernels polynomial; the kinetic energy uses the exact s-state radial Laplacian, whose lowest surviving power is the r^{-1} floor the auxiliary form supports. Two independent routes (explicit Laplacian vs the hydrogenic eigen-trick) agree *symbolically—difference exactly* 0—on homonuclear and heteronuclear sets. *Second*, the exchange assembly is re-run keeping every factor symbolic (the library entry point casts to float, capping certification at $\sim$16 digits).

[**MEASURED**] H_2 ($1s$ per center, $\zeta = 1$, $R = 1.4$): the homonuclear exchange Neumann τ-series *terminates symbolically* [**SYMBOLIC**] ($\tau = 1$ and all $\tau > 2$ are exact zeros), so the pipeline contains no truncation of any kind, and the FCI total energy is certified to **60 digits** (under the reference artifact's capping convention—a claim never exceeds the weaker run's requested precision; the dps 60 vs 90 evaluations in fact agree to 6×10^{-85}, so the recorded value is conservative):

$$E(1.4,\, \zeta{=}1) = -1.10655660609135850801940122475993772281832890689690719549979\ldots$$

with $R = 1.6$, 2.0 and the variational $\zeta = 1.197$ point certified alongside. Every integral matches an *independent* literature closed form to $\sim 10^{-61}$—the exchange integral against Sugiura [1], whose expression independently carries ln, Euler's γ and Ei, corroborating this paper's $\{E_1, \ln, \gamma\}$ seed set for that class from a route nearly a century removed—and matches direct quadrature at 10^{-21}–10^{-23}.

[**MEASURED**] LiH (s-only minimal: Li $1s, 2s$ + H $1s$, $R = 3.015$): the heteronuclear τ-series does *not* terminate, so the assembly is zero-quadrature but not truncation-free; the certified count is taken *net of an explicit tail bound* (measured monotone τ-ratios give $|\text{tail}| \leq 1.95 \times 10^{-32}$ summed over quartets, propagated through the 2-RDM weights and $\|S^{-1/2}\|^4$ to 1.2×10^{-30} Ha): $E_{\text{tot}} = -7.872619792415612173175580858 35\ldots$ Ha, **30 digits**. Honest scope, per the framework's own doctrine: these are *minimal-basis* numbers—the certified ζ=1 H_2 value sits 0.068 Ha above exact (0.027 Ha at the variational $\zeta = 1.197$ point) and the s-only LiH far above -8.0705—and what is certified is the assembly and its precision, not accuracy; the accuracy axis remains basis size. Certified values, with per-integral transcendence tags, are collected in the reference artifact (`docs/certified_reference_values.md`, `benchmarks/certified_reference/`).

[**SYMBOLIC + MEASURED**] The closed forms compose one level further: the *entire minimal-basis H_2 potential curve is a single explicit expression.* In the $1s/1s$ basis the singlet-Σ_g^+ CI space is $\{\, |g^2\rangle, |u^2\rangle \,\}$, so the FCI ground state is the quadratic-formula root of a 2×2 matrix whose entries are the closed-form molecular-orbital integrals,

$$E(R) = \tfrac{1}{2}(H_{11} + H_{22}) - \sqrt{\tfrac{1}{4}(H_{11} - H_{22})^2 + K_{gu}^2} + \frac{1}{R}, \tag{7}$$

with $H_{11} = 2h_{gg} + J_{gg}$, $H_{22} = 2h_{uu} + J_{uu}$, and every ingredient an explicit closed form in R—the assembled expression contains exactly the atoms $\{\exp, E_1, \ln, \gamma_E\}$ and nothing else beyond the rational and normalization prefactors of the opening list. A potential curve as a formula, not a point table: its exact R-derivatives certify the equilibrium constants by closed-form Newton iteration (`geovac/qfd_assemble.py::h2_closed_form_E`; `tests/test_paper58_qfd.py`),

$$R_{\text{eq}} = 1.66799996697274872492704641030726451334980414$$
$$D_e = 0.11865036209829531185349776433388467169190968$$
$$k = E''(R_{\text{eq}}) = 0.25470393430796998115272793572435567267825178$$

(Newton at dps 45 vs 60 agreeing to 9.5×10^{-46}; the curve dissociates to exactly -1 Ha—two ζ=1 hydrogen atoms—verified at 3×10^{-49}, so $D_e = -1 - E(R_{\text{eq}})$ is itself closed-form). Honest frame: these are the minimal-basis constants (R_{eq} exact is 1.401, k maps to $\omega_e \approx 3.17 \times 10^3$ cm^{-1} vs the measured 4401—the conversion importing the reduced mass is Layer-2; and the minimal-basis H_2 CI itself is as old as Sugiura's integral); what is new is their *status*: equilibrium geometry and force constant of a two-electron molecule as certified roots of one closed-form expression.

Two engine findings from the build. (i) *Domain restriction* (measured): the shell-route hybrid evaluator with $l_a, l_b > 0$ requires $Z_B < Z_A$ strictly—at $Z_B = Z_A$ a removable rate coincidence produces NaN (the limit exists and matches quadrature), and $Z_B > Z_A$ reaches an unimplemented Ei branch; a coverage gap, not a wrong value. (ii) A pre-existing exploratory driver's "DISAGREE beyond the fit floor—investigate" verdict against a Gaussian reference is resolved *against the reference*: the 8×10^{-6} gap sits in the Gaussian one-electron block (the cusp), while the native block matches literature closed forms to 60 digits.

VIII. SCOPE

What this paper does not claim. It does not claim the framework's atomic-sector results are affected: single-center angular structure is untouched, exact, and remains the basis of the atomic qubit results. It does not claim the qubit-resource results of Paper 14 are wrong; it clarifies that they are measured on the builder, whose neglected overlap is the subject here. And it does not claim a route around the obstruction: the census closes the one route the roadmap had left—untransformed bare-tensor use—while Observation 2 records only that the five relocations attempted so far each moved the cost without reducing it.

A distinction worth naming explicitly, because it bounds any scaling ambition. *Symmetry sparsity*—exact zeros forced by conservation laws, knowable from labels—is what this framework has, and Theorem 1 shows it is an atomic-sector property that degrades to an abelian residue at two centers and to a finite grading when linearity is lost. *Distance sparsity*—approximate near-zeros from spatial locality, exploited by screening, density fitting, and density-matrix truncation—is what linear-scaling electronic structure runs on, and it *improves* with system size. The framework has the first and none of the second. It is, structurally, a small-system framework, and both of its advantages contract as atom count grows: the abelian grading shrinks with symmetry, while the dense cross-center classes grow with the number of center pairs.

[1] Y. Sugiura, "Über die Eigenschaften des Wasserstoffmoleküls im Grundzustande," Z. Phys. **45**, 484–492 (1927).

[2] J. Loutey, "The Molecular Fock Projection: Diatomic Molecules as Prolate Spheroidal Lattices," GeoVac Paper 11 (2026).

[3] J. Loutey, "Structurally Sparse Qubit Hamiltonians from Spectral Graph Theory," GeoVac Paper 14 (2026).

[4] J. Loutey, "Composed Natural Geometries for Core-Valence Diatomics: LiH and BeH^+ from Discrete Graph Topology," GeoVac Paper 17 (2026).

[5] J. Loutey, "Toward PK-Free Molecular Hamiltonians: Two-Center Sturmian Integrals and Coupled Composition," GeoVac Paper 19 (2026).

[6] J. Loutey, "Structurally Sparse Molecular Hamiltonians for Near-Term Quantum Simulation," GeoVac Paper 20 (2026).

[7] K. P. Huber and G. Herzberg, *Constants of Diatomic Molecules* (Van Nostrand Reinhold, New York, 1979); ^{23}NaH $X\,^1\Sigma^+$ $r_e = 1.8874$ Å $= 3.5667\,a_0$, as tabulated in the NIST Chemistry WebBook (accessed 2026-08-11).

[8] R. S. Mulliken, C. A. Rieke, D. Orloff, and H. Orloff, "Formulas and numerical tables for overlap integrals," *J. Chem. Phys.* **17**, 1248–1267 (1949).

[9] C. C. J. Roothaan, "A study of two-center integrals useful in calculations on molecular structure. I," *J. Chem. Phys.* **19**, 1445–1458 (1951).

[10] K. Ruedenberg, "A study of two-center integrals useful in calculations on molecular structure. II. The two-center exchange integrals," *J. Chem. Phys.* **19**, 1459–1477 (1951).

[11] P.-O. Löwdin, "On the non-orthogonality problem connected with the use of atomic wave functions in the theory of molecules and crystals," *J. Chem. Phys.* **18**, 365–375 (1950).

[12] L. E. McMurchie and E. R. Davidson, "One- and two-electron integrals over Cartesian Gaussian functions," *J. Comput. Phys.* **26**, 218–231 (1978).

[13] E. Artacho and L. Miláns del Bosch, "Nonorthogonal basis sets in quantum mechanics: Representations and second quantization," *Phys. Rev. A* **43**, 5770–5777 (1991).

[14] J. Avery, *Hyperspherical Harmonics: Applications in Quantum Theory* (Kluwer Academic, Dordrecht, 1989).

[15] J. E. Avery and J. S. Avery, *Generalized Sturmians and Atomic Spectra* (World Scientific, Singapore, 2006).

[16] T. Shibuya and C. E. Wulfman, "Molecular orbitals in momentum space," *Proc. R. Soc. Lond. A* **286**, 376 (1965).

[17] P.-O. Löwdin, "Quantum theory of many-particle systems. I. Physical interpretations by means of density matrices, natural spin-orbitals, and convergence problems in the method of configurational interaction," *Phys. Rev.* **97**, 1474–1489 (1955).

[18] A. J. W. Thom and M. Head-Gordon, "Hartree–Fock solutions as a quasidiabatic basis for nonorthogonal configuration interaction," *J. Chem. Phys.* **131**, 124113 (2009).

[19] E. J. Sundstrom and M. Head-Gordon, "Non-orthogonal configuration interaction for the calculation of multielectron excited states," *J. Chem. Phys.* **140**, 114103 (2014).

[20] M. P. Barnett and C. A. Coulson, "The evaluation of integrals occurring in the theory of molecular structure. Parts I & II," *Philos. Trans. R. Soc. Lond. A* **243**, 221–249 (1951).

[21] E. Filter and E. O. Steinborn, "Extremely compact formulas for molecular two-center one-electron integrals and Coulomb integrals over Slater-type atomic orbitals," *Phys. Rev. A* **18**, 1 (1978).

[22] J. von Neumann and E. Wigner, "Über das Verhalten von Eigenwerten bei adiabatischen Prozessen," *Phys. Z.* **30**, 467–470 (1929).

[23] G. Herzberg, *Molecular Spectra and Molecular Structure. I. Spectra of Diatomic Molecules*, 2nd ed. (Van Nostrand, New York, 1950), Ch. VI (correlation of molecular and atomic states; the non-crossing rule).

[24] J. Loutey, "The Generalized-Sturmian Secular Equation Block-Encodes Without a Metric: A Quantum-Algorithm Analysis, Metric-Free for Atoms and a Conditioning Frontier for Molecules," GeoVac Paper 60 (2026).

[25] J. Loutey, "Bond Sphere: Diatomic Molecules as S^3 and their Natural Sturmian Basis," GeoVac Paper 8–9 (2026).

[26] A. T. Amos and G. G. Hall, "Single determinant wave functions," *Proc. R. Soc. Lond. A* **263**, 483 (1961).

[27] H. F. King, R. E. Stanton, H. Kim, R. E. Wyatt, and R. G. Parr, "Corresponding orbitals and the nonorthogonality problem in molecular quantum mechanics," *J. Chem. Phys.* **47**, 1936 (1967).

[28] A. C. West, M. W. Schmidt, M. S. Gordon, and K. Ruedenberg, "A comprehensive analysis of molecule-intrinsic quasi-atomic, bonding, and correlating orbitals. I. Hartree–Fock wave functions," *J. Chem. Phys.* **139**, 234107 (2013).

[29] P. R. Halmos, "Two subspaces," *Trans. Amer. Math. Soc.* **144**, 381 (1969).

[30] A. Böttcher and I. M. Spitkovsky, "A gentle guide to the basics of two projections theory," *Linear Algebra Appl.* **432**, 1412 (2010).

[31] T. P. Grozdanov and E. A. Solov'ev, "Separated- and united-atom limits for dynamical adiabatic states," *Phys. Rev. A* **44**, 5605 (1991).

[32] E. A. Solov'ev, "The advanced adiabatic approach and inelastic transitions via hidden crossings," *J. Phys. B: At. Mol. Opt. Phys.* **38**, R153 (2005).

[33] F. T. Smith, "Atomic distortion and the combining rule for repulsive potentials," *Phys. Rev. A* **5**, 1708 (1972).

[34] H.-Y. Huang, T.-L. Lu, T.-J. Whang, Y.-Y. Chang, and C.-C. Tsai, "Dissociation energy of the ground state of NaH," *J. Chem. Phys.* **133**, 044301 (2010).

TABLE IV. Claim → backing. LIVE = passing test in the `tests/test_paper58_*` suite (six files, plus `tests/test_two_center_eri_aabb.py` for the $(AA|BB)$ rows; per-claim mapping in `docs/claim_test_matrix.md`). OWED = result stands on an exploratory driver and must be migrated self-contained into `tests/` before this paper is a finished artifact. CORROB. = numerically corroborated but not exactly decided; see Sec. VIII.

claim	tier	backing
Table I S, h	MEASURED	LIVE
Table I g	COUNTED	CORROB.
2-center ERI engine, per class	MEASURED	LIVE
axial invariance	MEASURED	LIVE
Eq. (1)	SYMBOLIC	LIVE
Thm. 1(i)	INT.	LIVE
Thm. 1(ii)	INT.	LIVE
2-center machinery	SYMB.	LIVE
Obs. 1	MEASURED	LIVE
Pred. 1	UNTESTABLE	(needs angular l
Obs. 2	OBSERVATION	CHANGELOG
Correlation-diagram reading	OBSERVATION	Thm. 1(i)
Sec. IV front, exponent	MEASURED	LIVE
Eq. (5), t_c	MEASURED / SYMB.	LIVE
Obs. 3	MEASURED	LIVE
Tab. III, fixed R	MEAS.	LIVE
Tab. III, minima	MEAS.	LIVE
NaH atom-only ζ	MEAS.	LIVE
Obs. 4	MEASURED	LIVE
fit quality, span id.	MEASURED	LIVE
QFD 1e closed forms (Sec. VII)	SYMBOLIC	LIVE
QFD H_2 60-digit energy	MEASURED	LIVE (fast legs)
QFD LiH 30-digit (τ-tail net)	MEASURED	OWED (campaig

[35] C.-C. Chu, H.-Y. Huang, T.-J. Whang, and C.-C. Tsai, "Observation of double-well potential of NaH $C\,^1\Sigma^+$ state: deriving the dissociation energy of its ground state," *J. Chem. Phys.* **148**, 114301 (2018).

[36] F. Prosser and S. Hagstrom, "On the rapid computation of matrix elements," *Int. J. Quantum Chem.* **2**, 89 (1968).

[37] H. G. A. Burton, "Generalized nonorthogonal matrix elements: Unifying Wick's theorem and the Slater–Condon rules," *J. Chem. Phys.* **154**, 144109 (2021).

[38] P. Curie, "Sur la symétrie dans les phénomènes physiques, symétrie d'un champ électrique et d'un champ magnétique," *J. Phys. Théor. Appl.* **3**, 393 (1894).

[39] H. Bethe, "Termaufspaltung in Kristallen," *Ann. Phys. (Leipzig)* **395**, 133 (1929).

[40] R. S. Mulliken, "The interpretation of band spectra. Parts I, IIa, IIb," *Rev. Mod. Phys.* **2**, 60 (1930).

[41] R. S. Mulliken, "The interpretation of band spectra. Part III. Electron quantum numbers and states of molecules and their atoms," *Rev. Mod. Phys.* **4**, 1 (1932).

[42] C. Herring, "Critique of the Heitler–London method of calculating spin couplings at large distances," *Rev. Mod. Phys.* **34**, 631 (1962).

[43] C. Herring and M. Flicker, "Asymptotic exchange coupling of two hydrogen atoms," *Phys. Rev.* **134**, A362 (1964).

[44] NIST Digital Library of Mathematical Functions, §30.2–30.3 (spheroidal wave functions), `https://dlmf.nist.gov/30`.